

The big picture

Guiding question(s)

How are chemical equations used to calculate reacting ratios?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Cooking and chemistry may seem like two very different fields, but they actually have quite a lot in common. The same way that a chemist can work with different reactants to create a product, a chef also uses different ingredients to create a delicious meal. In fact, the process of cooking can be thought of as a series of chemical reactions that take place in the kitchen.

Ingredients used in cooking are similar to reactants in chemistry. As you will have noticed, the end product of cooking is quite different from the individual ingredients that make it up. For example, flour, sugar, and eggs on their own are not very appetising, but when combined and baked according to a recipe, they can create a delicious cake (**Figure 1**). What would happen if a chef accidentally used twice the amount of flour the recipe recommended? Or twice the number of eggs?

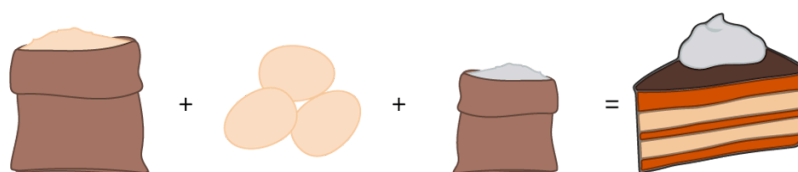


Figure 1. Recipes are similar to chemical equations.

In addition to combining ingredients, cooking also involves the careful consideration of ratios. Just like a chemist must balance the equation for a reaction, a cook must balance the ingredients in a recipe to achieve the desired flavour and texture. Adding too much or too little of an ingredient can completely change the outcome of a recipe, just as altering the ratios of reactants can also completely change the products which form.

Once the correct ratio of ingredients has been determined, a chef can also decide to increase the quantity of food produced by doubling or tripling the recipe. A chemist can do something similar – by keeping the correct ratio of reactants but

doubling their quantity double the amount of product created (**Table 1**).

Table 1. Cooking vs. chemistry.

Cooking	Chemistry
Uses specific ingredients to create a meal.	Uses specific reactants to create a product.
Ingredients must be in the correct ratio.	Reactants must be in the correct ratio.
Number of meals produced can be doubled by doubling the ingredients.	Amount of product can be doubled by doubling the reactants.

This subtopic will explore chemical equations and how the masses, concentration or volumes of reactants determine the quantity of product, as well as what quantity of reactants will remain unused. We will also explore different metrics for quantifying the success of these reactions.

To succeed in this subtopic of chemical equations and reacting ratios you will need to be secure in your understanding of the following areas of chemistry:

Prior learning

Before you study this subtopic make sure that you understand the following:

- States of matter and changes of state (see [section S1.1.2a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>)).
- What is meant by the term 'mole' (see [section S1.4.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-mole-and-avogadro-constant-id-43123/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-mole-and-avogadro-constant-id-43123/>)).
- How to convert between moles, molar mass (see [section S1.4.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/>)), molar concentrations (see [section S1.4.5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>)), and moles in gases (see [section S1.4.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/>)).

Practical skills

Once you have completed this subtopic, you can use your laboratory skills by going to [Practical 7: Synthesis of aspirin](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/synthesis-of-aspirin-id-46723/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/synthesis-of-aspirin-id-46723/>).